Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

"To reduce concern of all seats half torque in U321 model at Scorpio TCF by 90%"

Mahindra and Mahindra Ltd, Nashik Plant, Multi Model TCF.

	Plan	Actual
Start Date	**01.02.2019**	**01.02.2019**
Completion Date	**30.05.2019**	**30.11.2019**

Current Status: **Closed**

Cross Functional Team

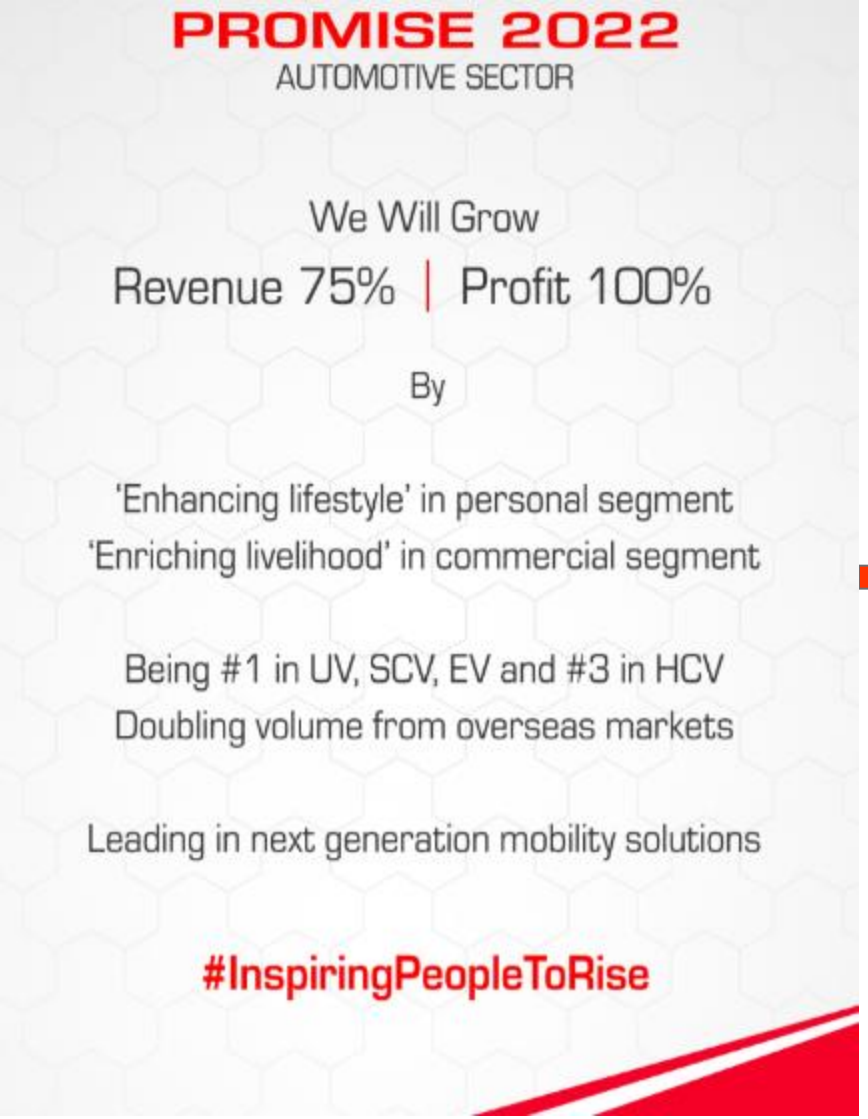
Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Shreeyash Nage	L8-O	STCF	9167382673
2	**Mentor**	Mandar Joshi	L5-M	STCF	9689003630
2	**Member 3**	Arvind Mistry	L7-M	PVT	9922957031
3	**Member 1**	Kiran Badgujar	L7-O	SQE	9689922687
4	**Member 2**	Keshav Badane	L7-O	SQE	9881355655
5	**Member 3**	Rahul Deore	L8-O	Bodyshop	9588432656
6	**Member 4**	Dnyaneshwar Wagh	L8-O	STCF	9730015208

Step1. Select Subject

Brief Description :

- What : Seat Bolt half torque
- Where : At trim-3 line and Final line after Seat assembly.
- When : Since Start of production of U321 vehicle i.e. from Sept-2018.
- Who : By stage Associate, Buy off check man.
- Why : Repeat issue in FAI & CAI.
- How : Avg 95 vehicles of per month.

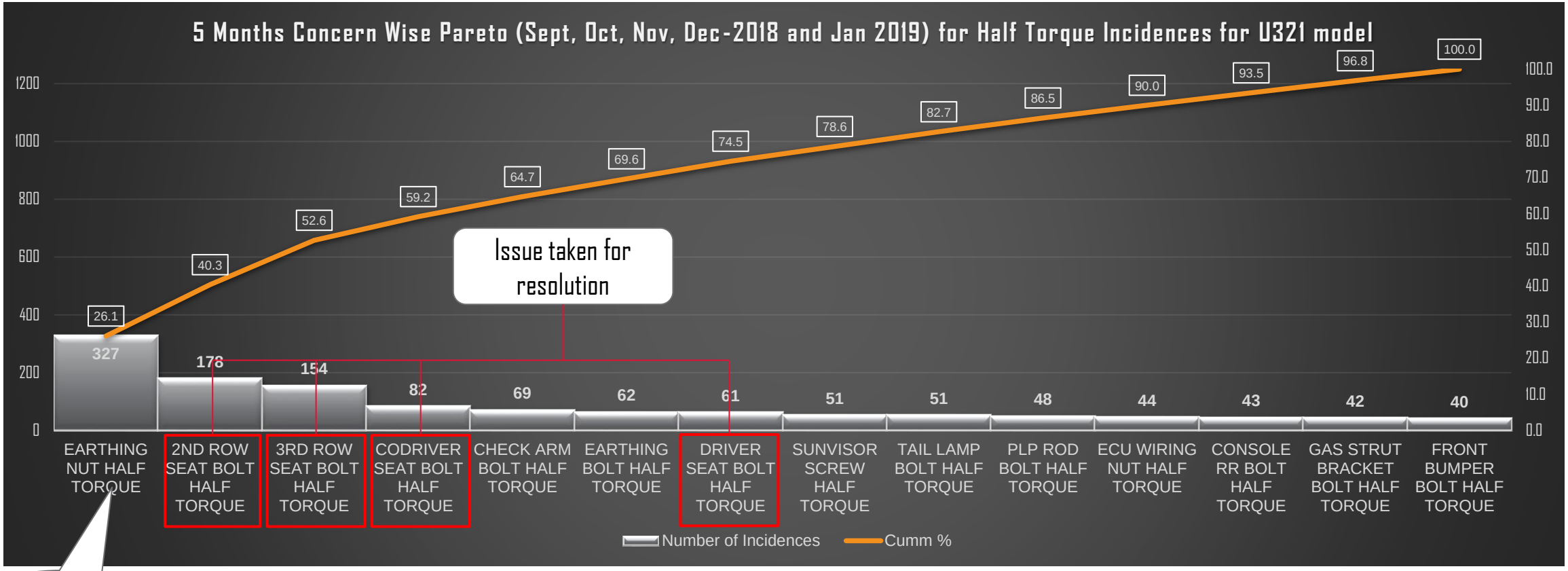
2. Justify the choice (reasons)



Crusade 2.0

Departmental Objective :-
– DPR Improvement
- **FRC / FBO Improvement**
- **Offline Rework Reduction**

2. Justify the choice



Defect selected for analysis by colleague

Inference:
Seat bolt half torque major contributor for incidences

2. Justify the choice

Cost of Poor Quality-Rework

Rework Cost estimation for seat bolt half torque-		
Description	Units	
Rework time (Approx)	Min	20
Avg Concerns/Day	Nos	3
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	193.55
Avg no. of working days	days	25
Operator cost /month	Rs	4838.71
Operator Cost/year	**Rs**	**58064.52**

Total Rework cost	Rs / Year	58065

2.b Immediate Containment Action

- ❑ Marking to be done on seat bolt after torqueing. Self certification by Stage associate.
- ❑ WIP checked.
- ❑ Check point added at Trim 3 buyoff and Final to check "seat bolt" independently.

Step 3. Understanding Current Situation

Mahindra

4M Analysis (Part 1)
(Conformity Analysis - 'L0')

PU/ Area : MultiModel TCF
Cell / BWT / Line: 5 and 11

Requested By- MQA | Date - 02.03.2019 | Source - Line | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement : 2nd Row, 3rd Row, Co-Driver and Driver Seat mounting Bolt half torque.

FMEA Severity /Gravity : ☑ V1+ ☐ V1 ☐ V2 ☐ V3
No of Incidences on the day (Including Yard check) : 3
VIN / Aggre. No.-
Model : Marazzo -U321

Date of Incidence/ Assy : 02.02.2019
Shift / Time : Both
Whether SC/CC/Z parameter affected ? ☑ YES ☐ NO ☐ Z ☑ CC ☐ SC

Immediate containment action for Customer Protection :
Yard Check Done.

Does a related 4M Analysis - Part 1 or QC story already exist ? ☑ YES ☐ NO | If Yes | 4M Analysis Part 1 Sr. No. | QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned- BWT . :

Does a control already exist ? ☑ YES ☐ NO

Stage / Post Analysis -

Man(Human Action)

Associate Name : B D Ahire
Is operator trained ? ☑ YES ☐ NO
Operator level (I.L.U) Black L
OR Dexterity level(1...4) L4
☐ YES ☑ NO
Is operator under training ? ☐ YES ☑ NO
Is operator temporary for Stage ? ☐ YES ☑ NO
Bad rework ? ☐ YES ☐ NO

Method(process)

Main steps are respected ? ☑ YES ☐ NO
Is there a related key point in SOP / SOS? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ? ☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ? ☑ YES ☐ NO

Other

Is there a Control Plan ? (Record in Analysis table) ☑ YES ☐ NO
Is the packaging standard defined for the part ? ☐ NO
Is the packaging st[illegible] respected ? ☐ YES ☑ NO
Is there a[illegible] mechan[illegible] for failure ? ☐ YES ☑ NO

Machine

Is the specified /Tool / Fixture facility used ? ☑ YES ☐ NO
Last calibration dates
Facility /Tool / Fixture in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters known ? (Record in Analysis table) ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Product Failure

In only one shift ? ☐ YES ☑ NO
In only one line ? ☐ YES ☑ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO
Is the suspected parameter measurable ? ☑ YES ☐ NO (Record in Analysis table)

Material/ Consumable

Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☑ NO
Is the storage SOP respected ? ☑ YES ☐ NO
Is the shelf life / expiry date respected ? ☑ YES ☐ NO
Is the part as per FIFO? ☑ YES ☐ NO

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
Bolt found half torque. Prefitment difficult as per feedback from associate.

Traveller card: There is remark written in traveller card of defect.

History: No defect history as new vehicle launch.

Detection: Self certification done be stage associate

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark
1	Tool Torque	45 +/- 5 Nm	45 Nm	Ok

Is 4M Analysis Part 2 required ? ☑ YES ☐ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action :-	Resp	Date	Status
DRF Given to stage associate & Buy off associate to identify the defect.	Shreeyash Nage	29.01.20	Closed

2. Permanent Action -	Resp	Date	Status
1. L1 analysis required.	Shreeyash Nage	21.02.20	Open

Analysed by Name/s - Shreeyash Nage
Date - 30.01.20
Choice : ☐ WSS ☑ QC Story

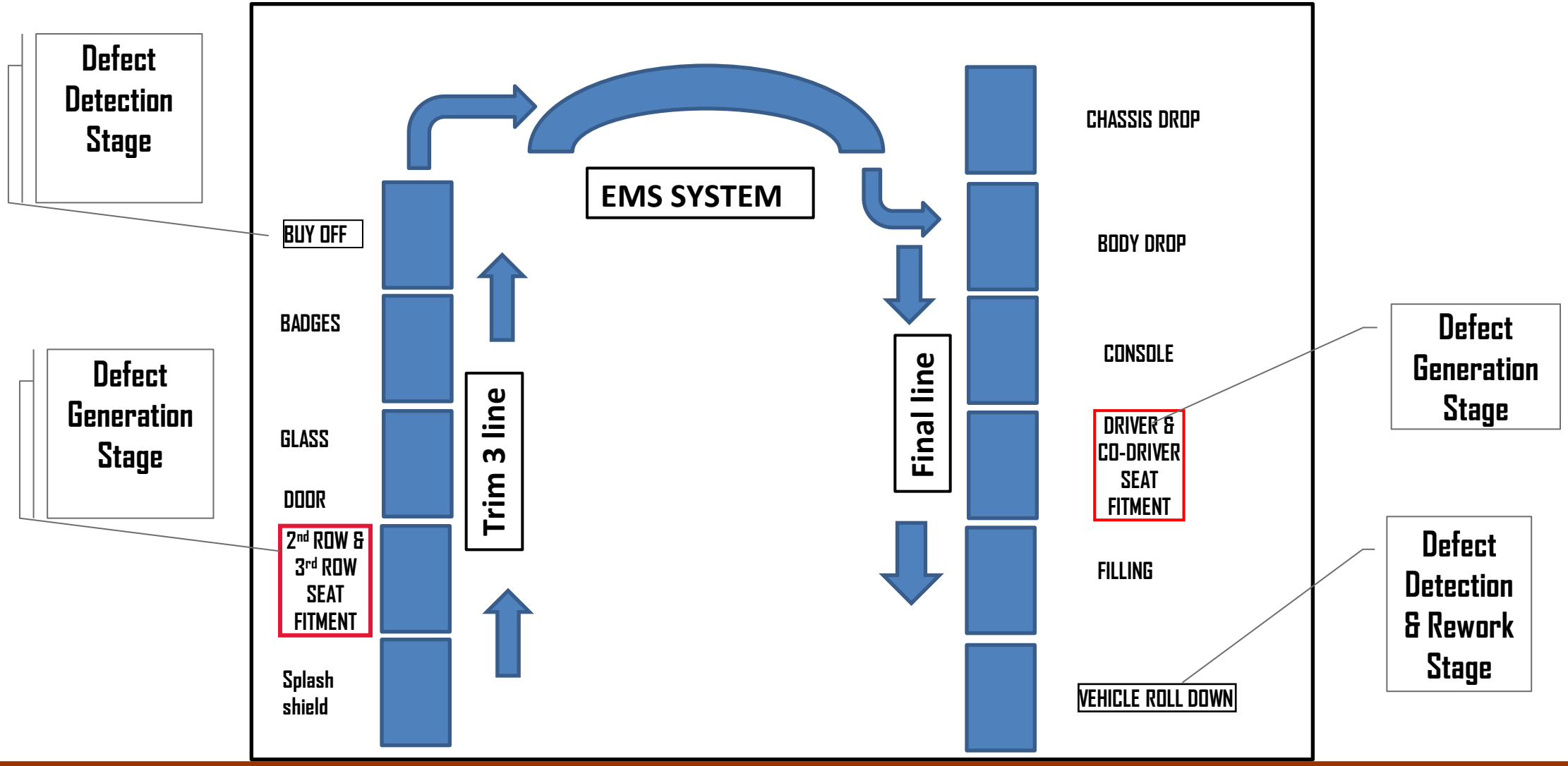
Inference from L0,
Problem follows part & vehicle
Problem is a repeat problem &
hence detailed analysis required.

NO QUICK WIN.

4M analysis Good and Not so good observations	
MAN	Operator Dext. Level L4
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	**Problem follows part**
Material	Handling and shelf life of part adhered

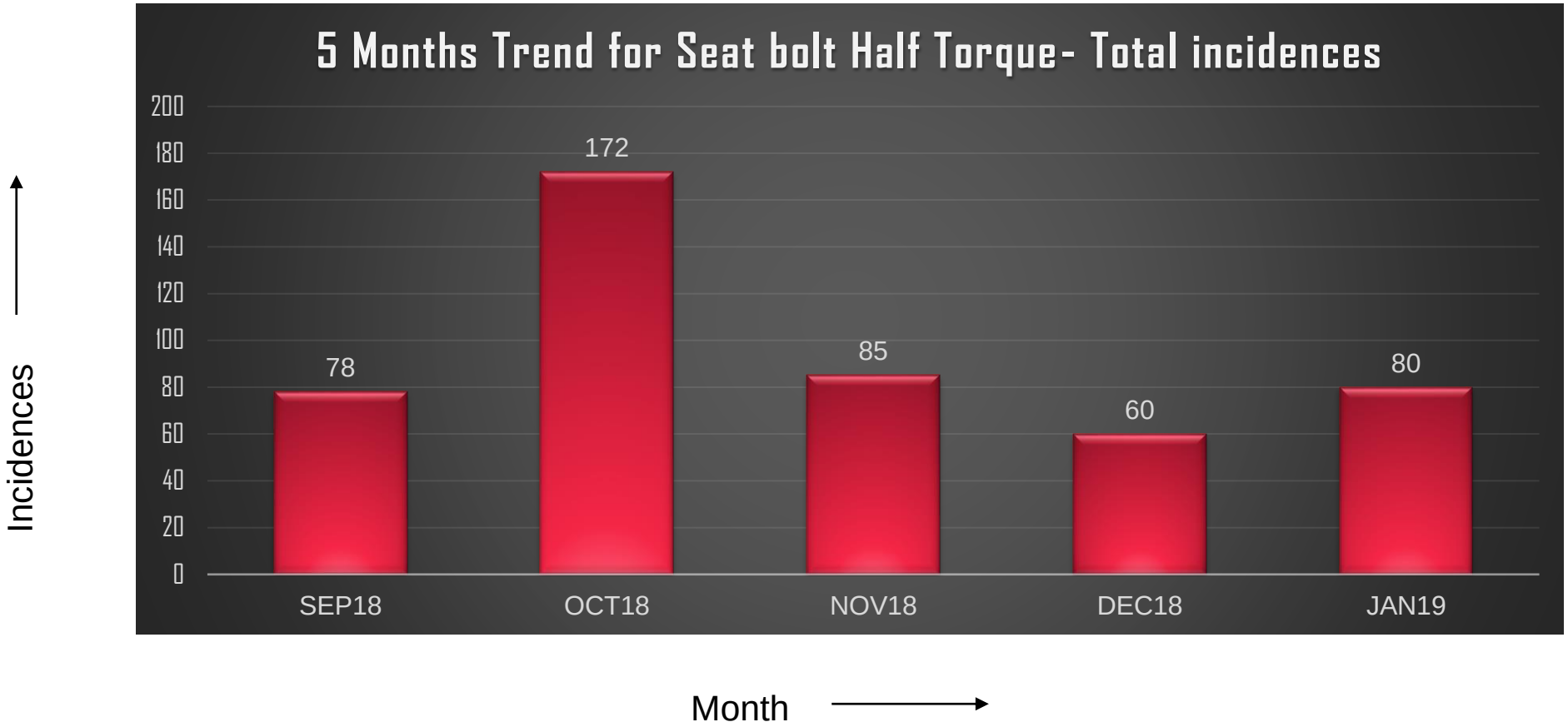
3. Understanding The Current Situation

OBSERVATION OF PROCESS : BODY FLOW DIRECTION FROM TRIM 3 LINE TO FINAL LINE



3. Understanding The Current Situation

Monthly defect trend



3. Understanding The Current Situation

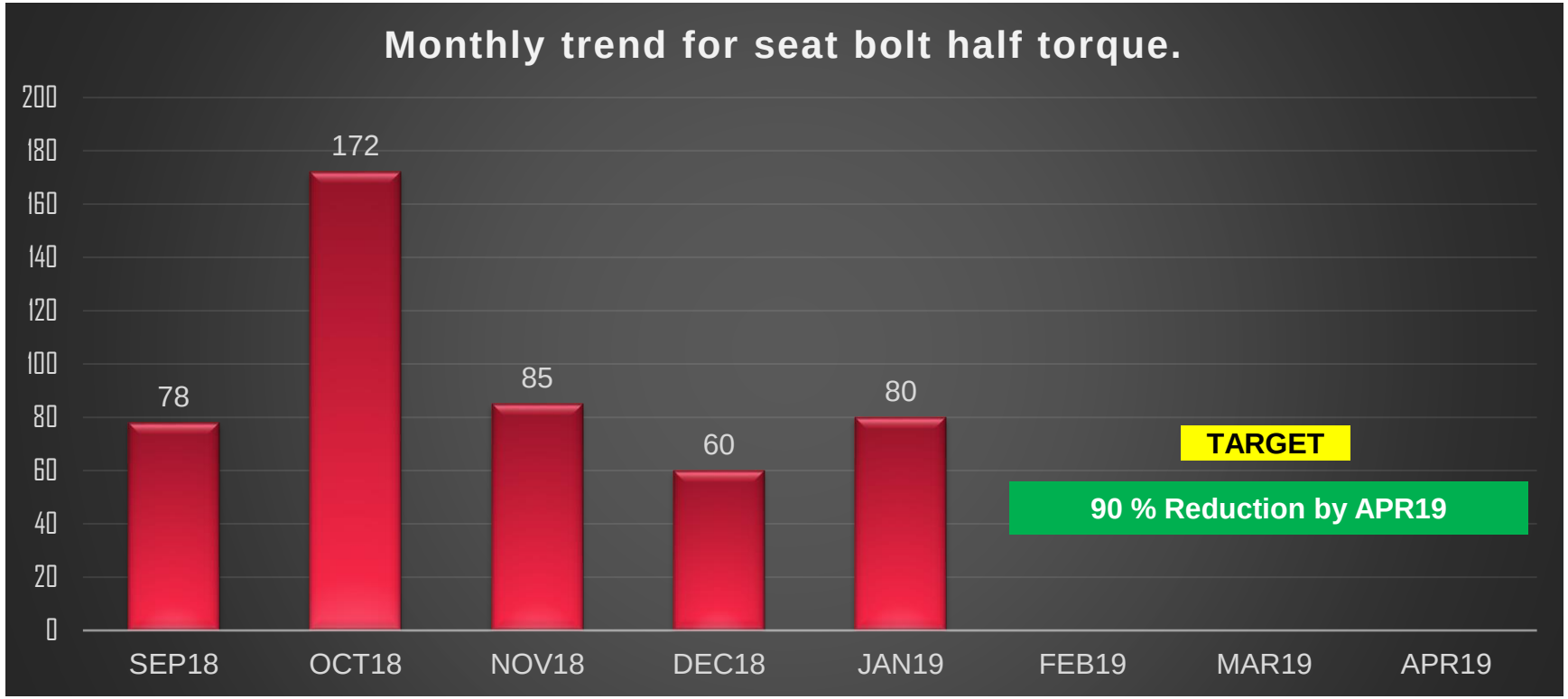
Good & Not so good observations

- Issue observed only in Marazzo (U321) Model.
- Issue reported in both the shifts.
- Repair process is tricky.
- Issue reported at stage.
- Not all the vehicles having this issue.
- Bolt need to change while removal from vehicle.
- Part getting rejected while rework of defect.
- During Seat removal and refitment activities other defects such as scratches and damages occur. Rework for same is added.

4. Select targets

Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	95 Nos Average monthly incidences
Target :	**90 % Reduction by APR19**



INFERENCE:- Target taken to reduce concern by April 2019.

Step 5. Analysis

List of Probable causes identification in Brain storming session

Sr.no	Possible cause	Sr.no	Possible cause
1	New operator deployed at stage.	8	Tool RPM not ok.
2	SOS Not Followed .	9	Welding Spatter in body nut.
3	Unskilled operator for seat assembly	10	Excess ED deposition in nut
4	Body nut plate offset	11	Seat and body alignment not okay
5	Bolt pitch not ok	12	Nut thread quality not okay.
6	Tool torque not ok.	13	Wrong Bolt used
7	Bolt axis not perpendicular to nut axis (not prefit properly)	14	Seat bracket bend(handling damage)

Step 5. Analysis

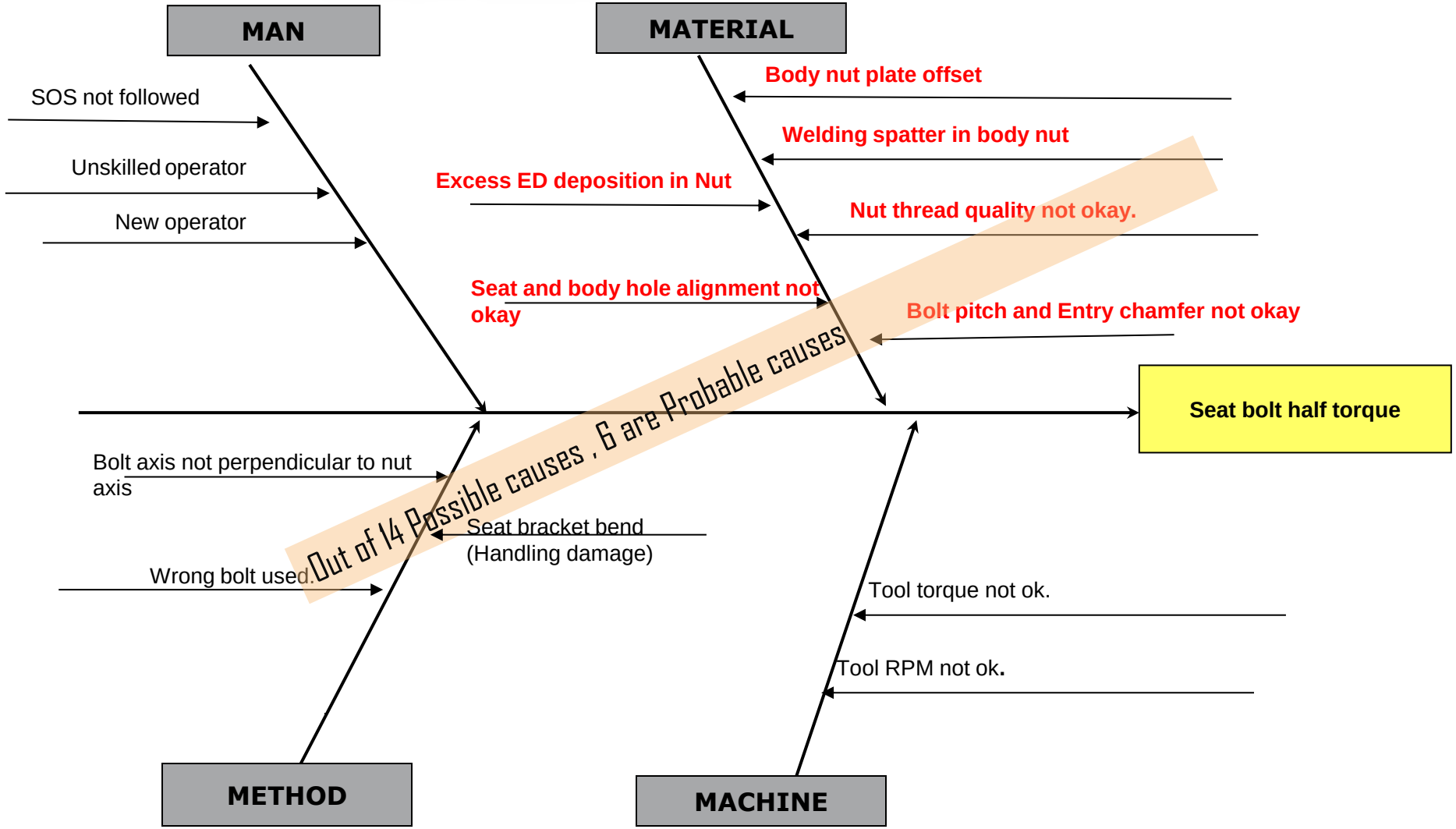
List of Probable causes identification in Brain storming session

Sr. no	Possible Causes	Shreeyash	Keshav	Kiran	Arvind	Avesh	Rahul	Hemant	Total
1	New operator deployed at stage.	1	1	3	3	3	3	3	**17**
2	SOS Not Followed .	1	1	1	1	1	1	1	**7**
3	Unskilled operator for seat assembly	3	3	3	3	1	3	3	**19**
4	Body nut plate offset	9	9	9	3	3	3	3	**39**
5	Bolt pitch not ok	9	9	9	3	1	1	1	**33**
6	Tool torque not ok.	1	3	1	3	3	1	3	**15**
7	Bolt axis not perpendicular to nut axis	9	3	3	3	9	3	3	**33**
8	Tool RPM not ok.	1	9	3	3	1	1	3	**21**
9	Welding Spatter in body nut.	9	3	3	3	9	3	3	**33**
10	Excess ED deposition in nut	3	3	1	3	3	3	9	**25**
11	Seat and body alignment not okay	9	3	3	3	3	9	9	**39**
12	Nut thread quality not okay.	3	9	9	3	9	1	3	**37**
13	Wrong Bolt used	1	3	3	3	3	1	3	**17**
14	Seat bracket bend(handling damage)	3	1	9	3	1	1	3	**21**

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 33.

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

List of causes not considering probable cause and reason for not considering

Sr. No	Not considered Causes	Reasons
1	New operator deployed at stage.	All trained operator deployed on stage
2	SOS Not Followed .	SOS Followed
3	Unskilled operator for seat assembly	All trained operator deployed on stage
6	Tool torque not ok.	Tool Torque Verified found okay
8	Tool RPM Not ok.	Tool RPM Verified found okay
10	Excess ED deposition in nut	Checked visually on by fitting bolt.
13	Wrong Bolt used	Color difference between Scorpio and Marazzo Bolt
14	Seat bracket bend(handling damage)	Handling checked found okay.

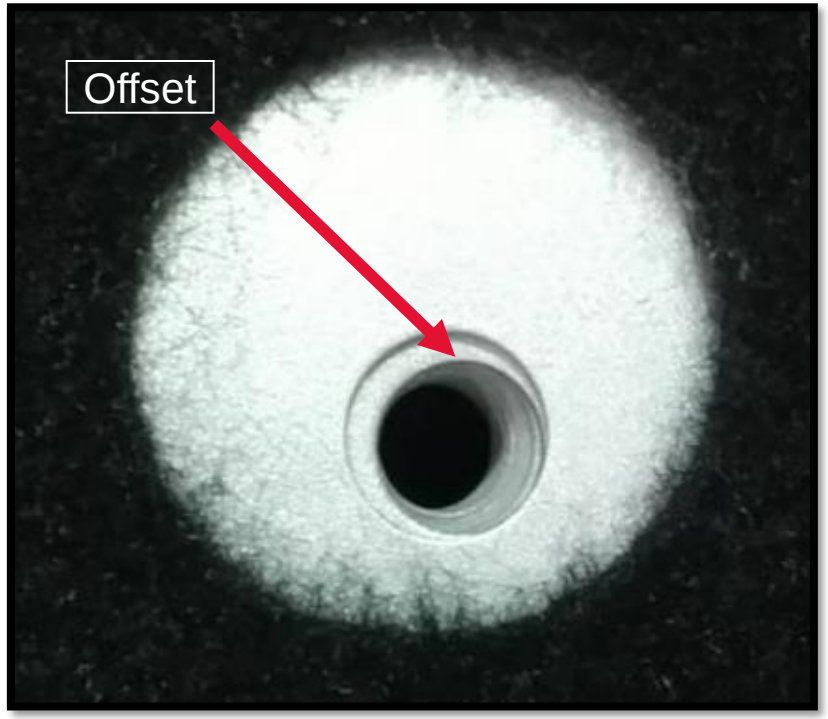
5.1 Identifying Probable cause

No	List of Probable Causes
4	Body nut plate offset.
5	Bolt pitch and entry chamfer not okay.
7	Bolt axis not perpendicular to nut axis. (not prefit properly)
9	Welding Spatter in body nut.
11	Seat and body alignment not okay.
12	Nut thread quality not okay.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Body nut plate offset.	L1 analysis report/ Online checking by fitting bolt at body IN.	Bodyshop representative deployed for checking body nut at body In at TCF. Checked 50 vehicles ,4 found offset. Bolt prefitment not possible.	**VALID**



Observations:-

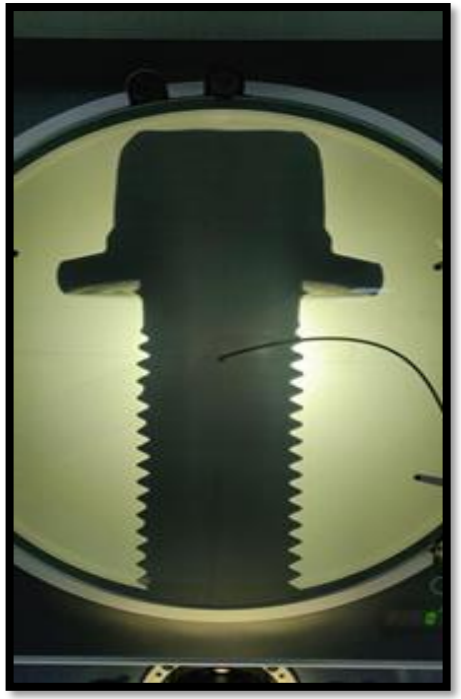
- Total 50 nos. of BIW checked by fitting bolt in nut for nut offset.
- First Row, second row and third row all nuts checked
- Out of this 50 nos., 4 vehicles found with welding spatter in **2nd Row and 3rd Row seat nut.**

Conclusion: Body nut plate offset contributing to seat bolt half torque.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Bolt pitch and entry chamfer not okay	L1 analysis report	Screw pitch is measured on profile projector – found ok.	**INVALID**



Observation :-

- 10 samples of seat bolt were check on Profile Projector. Pitch ,major and minor diameter checked.
- Found ok.

Conclusion:- Bolt pitch and diameter is okay. Not contributing to the defect.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Bolt axis not perpendicular to nut axis. (not prefit properly)	Online checking by fitting bolt at seat fitment stage.	50 vehicles checked for seat bolt prefitment along with seat mounted on body.	**VALID**

Observations:-

- Total 50 nos of BIW checked by fitting bolt in nut for nut offset.
- First Row, second row and third row all nuts checked
- Out of this 50 nos, 20 vehicles found with bolt axis not perpendicular to nut axis, ie not prefit properly in 1st Row, 2nd Row and 3rd Row seat.

Additional Observations:-

- Seat bolt of Marazzo Vehicle was compared with Scorpio.
- In spite, entry chamfer of Marazzo found okay, Dog point was missing as present in Scorpio seat bolt.

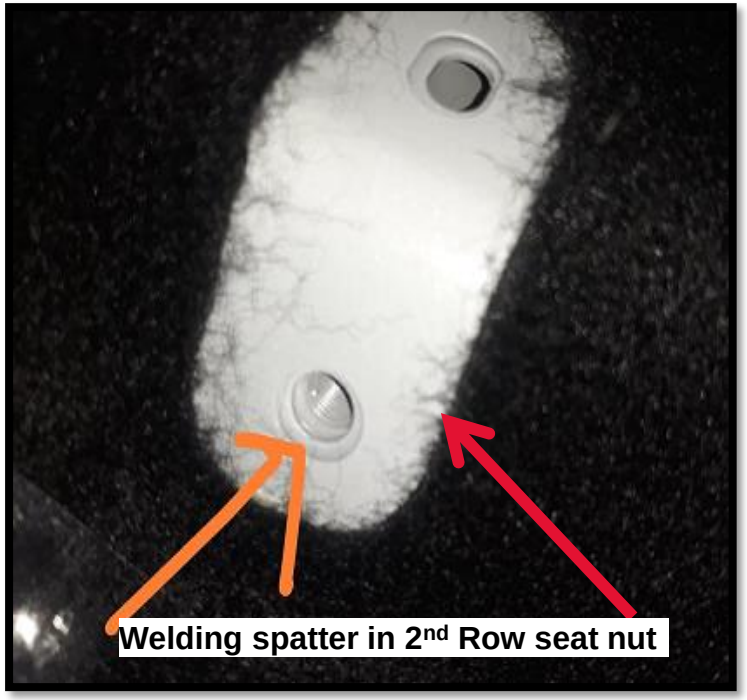
Conclusion:- In spite of trained operator on all three stages, bolt prefitment not done properly. Root cause for majority defect for all three stages lies that bolt not prefitted properly due to dog point missing on bolt.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
9	Welding Spatter in body nut.	Visual Inspection	50 vehicles checked for visually for welding spatter inside nut.	**VALID**

Welding spatter found only in 2nd row and 3rd row seat nut.



Observations:-

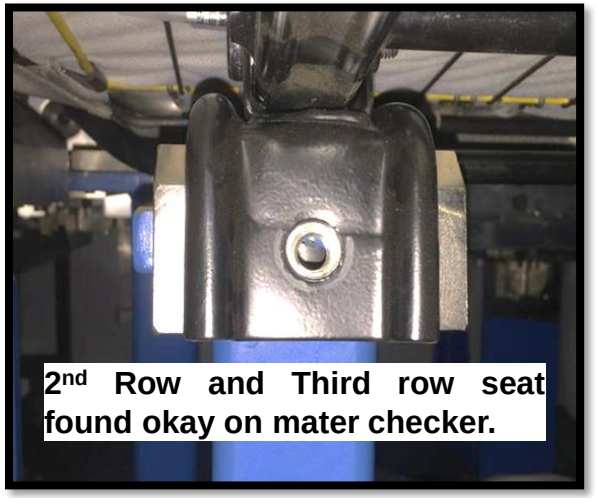
- Total 50 nos of BIW checked by fitting bolt in nut for and welding spatter presence..
- First Row, second row and third row all nuts checked
- Out of this 50 nos 3 vehicles found with welding spatter in 2nd Row and 3rd Row seat nut.

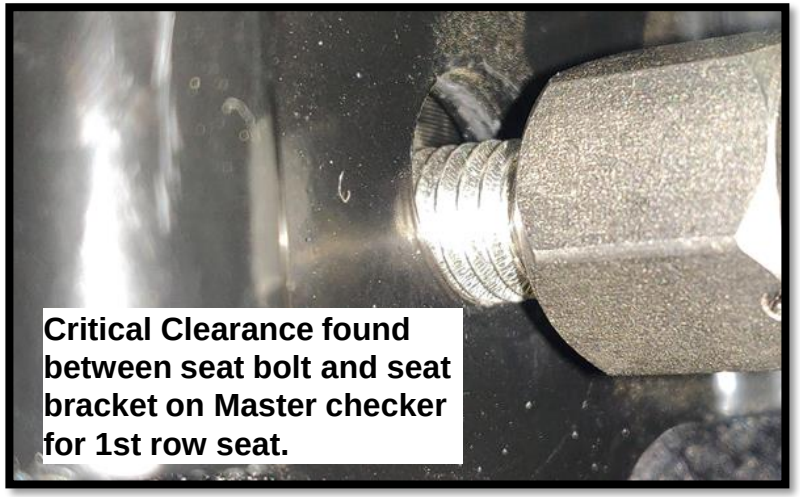
Conclusion:- Welding Spatter contributing in 2nd Row and 3rd Row seat bolt half torque.

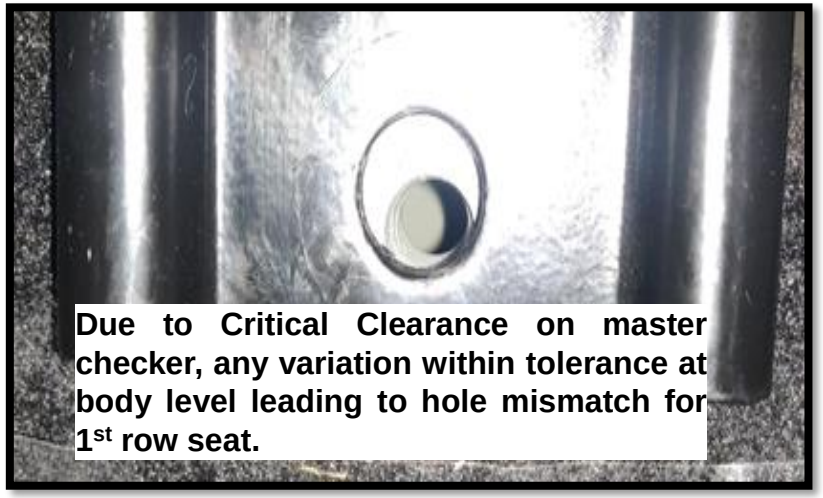
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
11	Seat bracket and body alignment not okay.	Master checker	5 samples of First Row, second row and third row seat were checked on master checker.	**VALID**







Observations:-

- Total 5 nos. of First Row, second row and third row all Seats checked Seat checked on Master checker for hole mismatch.
- 2nd Row and 3rd row seat hole found matching with Master checker.
- All 5 first row seat found with critical clearance between bolt and seat bracket.

Conclusion:- Seat bracket and body alignment not okay for **first row seat.**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
12	Nut thread quality not okay.	Online checking by fitting bolt at body in at Trim 1.	Bodyshop representative deployed for checking body nut at body In at TCF. Checked 50 vehicles ,4found offset. Bolt prefitment not possible.	**INVALID**

Observations:-

- Total 50 nos. of BIW checked by fitting bolt in nut for nut thread quality.
- First Row, second row and third row all nuts checked.
- Out of this 50 nos. no vehicles found with nut thread quality issue.

Conclusion: Thread quality found okay. Not contributing to defect.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Conclusion
4	Body nut plate offset.	L1 analysis report/ Online checking by fitting bolt at body IN.	**Hypothesis valid**
7	Bolt axis not perpendicular to nut axis. (not prefit properly)	Online checking by fitting bolt at seat fitment stage.	**Hypothesis valid**
9	Welding Spatter in body nut.	Visual Inspection	**Hypothesis valid**
11	Seat bracket and body alignment not okay.	Master checker	**Hypothesis valid**

4 Valid Causes

WHY - WHY Analysis

5.3 Identify the root causes:

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
4	Body nut plate offset.	Nut offset during welding	Excess Clearance in Locating pin diameter and Fixture hole	Uniform diameter in compliance with nut diameter present.	Variable diameter stepped pin to be provided during welding.
7	Bolt axis not perpendicular to nut axis. (not prefit properly)	Prefitment not possible	Dog point not present on bolt.	Design Flaw.	Dog point to be added on bolt.
9	Welding Spatter in body nut.	Spatter generated during $C0_2$ welding.	No protection on nut during welding		Protection to be provided during welding.
11	Seat bracket and body alignment not okay.	Critical clearance between bolt and bracket	Seat track checking fixture Dimension found not ok Required 26.6mm found 25.7mm	Track riveting fixture block worn out.	Track riveting fixture block to be replaced.

6. Implement the corrective measures

Possible Remedies Identified:

One possible solution for All seat bolt half torque

Legends	
High	Low

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Bolt prefitment not possible	Dock point to be added on bolt.	High	High	No	High	Accept

6. Implement the corrective measures

Possible Remedies Identified:

One possible solution for All seat bolt half torque

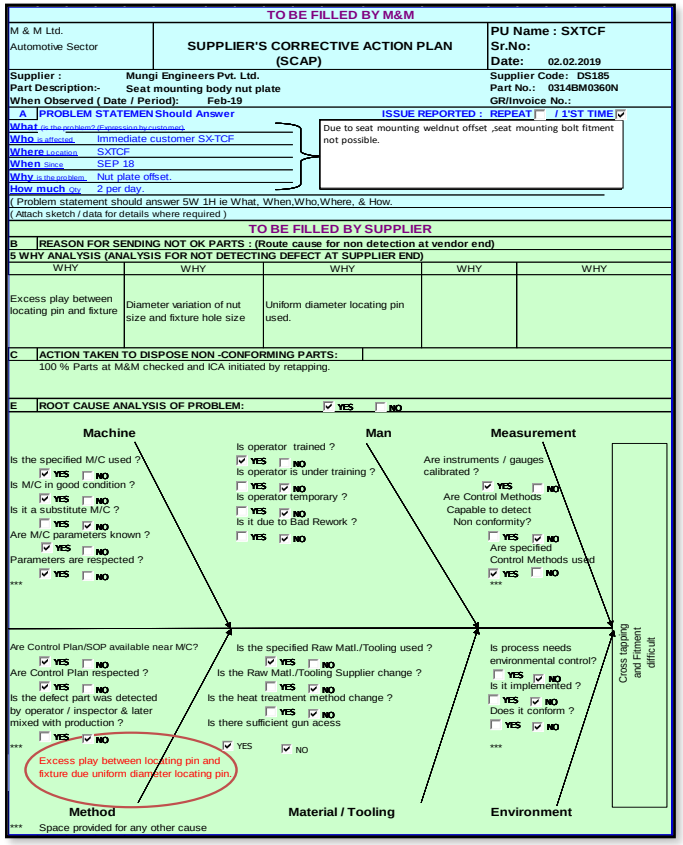
Legends	
High	Low

S N	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Critical clearance between bolt and seat bracket.	Correction to be done in track riveting fixture at supplier end.	High	High	No	High	Accept

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
4	Body nut plate offset	Variable diameter stepped pin to be provided during welding.	20.02.2019	Keshav Badane	CLOSED



TO BE FILLED BY M&M

M & M Ltd.
Automotive Sector

SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP)

PU Name : SXTCF
Sr.No:
Date: 02.02.2019

Supplier : Mungi Engineers Pvt. Ltd.
Part Description:- Seat mounting body nut plate
When Observed (Date / Period): Feb-19

Supplier Code: DS185
Part No.: 0314BM0360N
GR/Invoice No.:

A PROBLEM STATEMEN Should Answer — ISSUE REPORTED : REPEAT ☐ / 1'ST TIME ☑

What (is the problem? (Expressing by customer))
Who is affected. Immediate customer SX-TCF
Where Location SXTCF
When Since SEP 18
Why is the problem. Nut plate offset.
How much Qty 2 per day.

Due to seat mounting weldnut offset ,seat mounting bolt fitment not possible.

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)
5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Excess play between locating pin and fixture	Diameter variation of nut size and fixture hole size	Uniform diameter locating pin used.		

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:
100 % Parts at M&M checked and ICA initiated by retapping.

E ROOT CAUSE ANALYSIS OF PROBLEM: ☑ YES ☐ NO

Machine
Is the specified M/C used ? ☑ YES ☐ NO
Is M/C in good condition ? ☑ YES ☐ NO
Is it a substitute M/C ? ☐ YES ☑ NO
Are M/C parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO
***

Man
Is operator trained ? ☑ YES ☐ NO
Is operator is under training ? ☐ YES ☑ NO
Is operator temporary ? ☐ YES ☑ NO
Is it due to Bad Rework ? ☐ YES ☑ NO

Measurement
Are instruments / gauges calibrated ? ☑ YES ☐ NO
Are Control Methods Capable to detect Non conformity? ☐ YES ☑ NO
Are specified Control Methods used ☑ YES ☐ NO
***

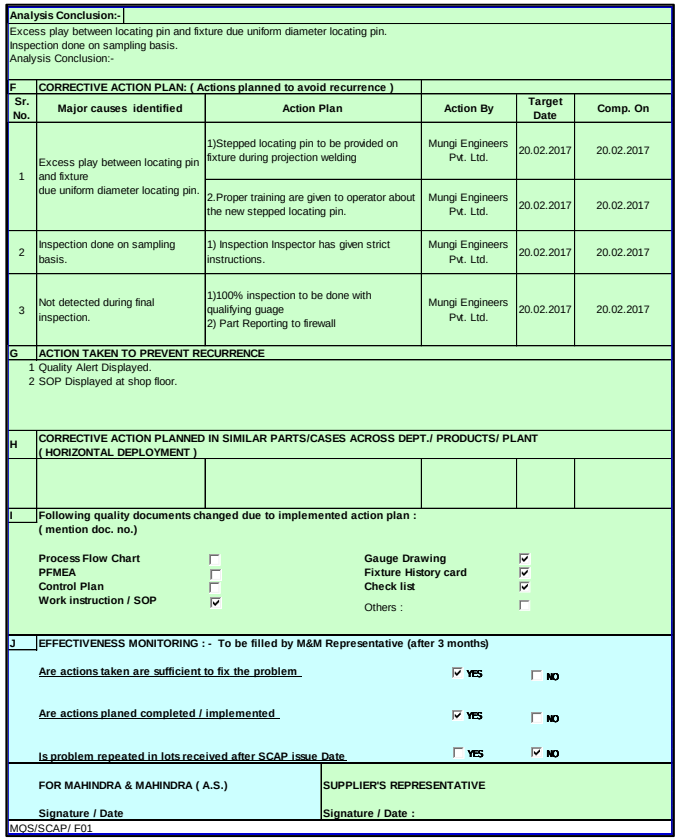
Cross tapping and Fitment difficult

Method
Are Control Plan/SOP available near M/C? ☑ YES ☐ NO
Are Control Plan respected ? ☑ YES ☐ NO
Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO
***
Excess play between locating pin and fixture due uniform diameter locating pin.

Material / Tooling
Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO
Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO
Is the heat treatment method change ? ☐ YES ☑ NO
Is there sufficient gun acess ☑ YES ☐ NO

Environment
Is process needs environmental control? ☐ YES ☑ NO
Is it implemented ? ☐ YES ☑ NO
Does it conform ? ☐ YES ☑ NO
***

*** Space provided for any other cause



Analysis Conclusion:-
Excess play between locating pin and fixture due uniform diameter locating pin.
Inspection done on sampling basis.
Analysis Conclusion:-

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	Excess play between locating pin and fixture due uniform diameter locating pin.	1)Stepped locating pin to be provided on fixture during projection welding	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
		2.Proper training are given to operator about the new stepped locating pin.	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
2	Inspection done on sampling basis.	1) Inspection Inspector has given strict instructions.	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
3	Not detected during final inspection.	1)100% inspection to be done with qualifying guage 2) Part Reporting to firewall	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017

G ACTION TAKEN TO PREVENT RECURRENCE
1 Quality Alert Displayed.
2 SOP Displayed at shop floor.

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

I Following quality documents changed due to implemented action plan : (mention doc. no.)

Process Flow Chart ☐
PFMEA ☐
Control Plan ☐
Work instruction / SOP ☑
Gauge Drawing ☑
Fixture History card ☑
Check list ☑
Others : ☐

J EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem ☑ YES ☐ NO
Are actions planed completed / implemented ☑ YES ☐ NO
Is problem repeated in lots received after SCAP issue Date ☐ YES ☑ NO

FOR MAHINDRA & MAHINDRA (A.S.) — Signature / Date
SUPPLIER'S REPRESENTATIVE — Signature / Date :

MOS/SCAP/ F01

OBSERVATIONS AND ACTIONS :-

Uniform diameter pin used for locating nut.

Difference in diameter on nut and fixture hole leading to excess play.

Stepped diameter pin provided to nullify the play.

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
7	Bolt axis not perpendicular to nut axis. (not prefit properly)	Dog point to be added on bolt.	PTR started -23.08.2019 (Implementation of new bolt done immediately after PTR)	Arvind Mistry	CLOSED

OBSERVATIONS AND ACTIONS :-

- Scorpio vehicle bolt compared with Marazzo seat bolt.
- Dog point found missing in Marazzo seat bolt.
- Same issue taken up with PVT for ECN change.

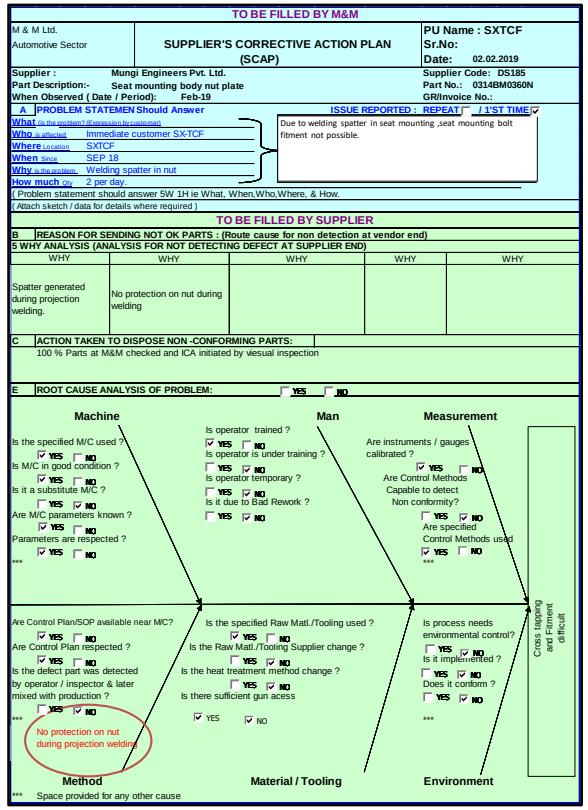
Marazzo without dog point

Recommended Bolt with dog point

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
4	Welding Spatter in body nut.	Protective fixture to be placed before welding.	20.02.19	Rahul Deore/Keshav Badane.	CLOSED



TO BE FILLED BY M&M

M & M Ltd. Automotive Sector | SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP) | PU Name : SXTCF, Sr.No:, Date: 02.02.2019

Supplier : Mungi Engineers Pvt. Ltd. | Supplier Code: DS185
Part Description:- Seat mounting body nut plate | Part No.: 0314BM0360N
When Observed (Date / Period): Feb-19 | GR/Invoice No.:

A PROBLEM STATEMEN Should Answer — ISSUE REPORTED : REPEAT ☐ 1 1ST TIME ☑

What (is the problem? Expression by customer) — Due to welding spatter in seat mounting ,seat mounting bolt fitment not possible.
Who is affected — Immediate customer SX-TCF
Where Location — SXTCF
When Since — SEP 18
Why is the problem — Welding spatter in nut
How much Qty — 2 per day.

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)
5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Spatter generated during projection welding.	No protection on nut during welding			

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:
100 % Parts at M&M checked and ICA initiated by visual inspection

E ROOT CAUSE ANALYSIS OF PROBLEM: ☐ YES ☐ NO

Machine: Is the specified M/C used ? ☑ YES ☐ NO; Is M/C in good condition ? ☑ YES ☐ NO; Is it a substitute M/C ? ☐ YES ☑ NO; Are M/C parameters known ? ☑ YES ☐ NO; Parameters are respected ? ☑ YES ☐ NO; ***

Man: Is operator trained ? ☑ YES ☐ NO; Is operator is under training ? ☐ YES ☑ NO; Is operator temporary ? ☐ YES ☑ NO; Is it due to Bad Rework ? ☐ YES ☑ NO

Measurement: Are instruments / gauges calibrated ? ☑ YES ☐ NO; Are Control Methods Capable to detect Non conformity? ☐ YES ☑ NO; Are specified Control Methods used ☑ YES ☐ NO; ***

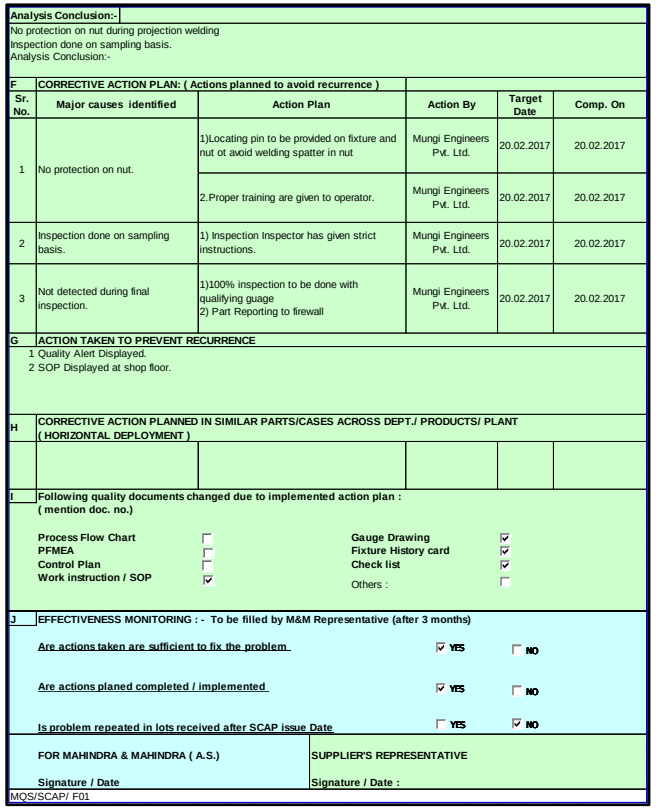
Method: Are Control Plan/SOP available near M/C? ☑ YES ☐ NO; Are Control Plan respected ? ☑ YES ☐ NO; Is the defect part was detected by operator / inspector & later mixed with production ? ☐ YES ☑ NO; ***; No protection on nut during projection welding

Material / Tooling: Is the specified Raw Matl./Tooling used ? ☑ YES ☐ NO; Is the Raw Matl./Tooling Supplier change ? ☐ YES ☑ NO; Is the heat treatment method change ? ☐ YES ☑ NO; Is there sufficient gun acess ☑ YES ☑ NO

Environment: Is process needs environmental control? ☐ YES ☑ NO; Is it implemented ? ☐ YES ☑ NO; Does it conform ? ☐ YES ☑ NO; ***

Cross Tagging area Fitment difficult

*** Space provided for any other cause



Analysis Conclusion:-
No protection on nut during projection welding
Inspection done on sampling basis.
Analysis Conclusion:-

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	No protection on nut.	1)Locating pin to be provided on fixture and nut ot avoid welding spatter in nut	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
		2.Proper training are given to operator.	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
2	Inspection done on sampling basis.	1) Inspection Inspector has given strict instructions.	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017
3	Not detected during final inspection.	1)100% inspection to be done with qualifying guage 2) Part Reporting to firewall	Mungi Engineers Pvt. Ltd.	20.02.2017	20.02.2017

G ACTION TAKEN TO PREVENT RECURRENCE
1 Quality Alert Displayed.
2 SOP Displayed at shop floor.

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

I Following quality documents changed due to implemented action plan : (mention doc. no.)

Process Flow Chart ☐ | Gauge Drawing ☑
PFMEA ☐ | Fixture History card ☑
Control Plan ☐ | Check list ☑
Work instruction / SOP ☑ | Others : ☐

J EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem ☑ YES ☐ NO
Are actions planed completed / implemented ☑ YES ☐ NO
Is problem repeated in lots received after SCAP issue Date ☐ YES ☑ NO

FOR MAHINDRA & MAHINDRA (A.S.) — Signature / Date
SUPPLIER'S REPRESENTATIVE — Signature / Date :

MQS/SCAP/ F01

OBSERVATIONS AND ACTIONS :-

- Welding spatter generated during CO_2 welding in Bodyshop.
- Similar observation noted at supplier end.
- Check point added and sampling inspection plan made.

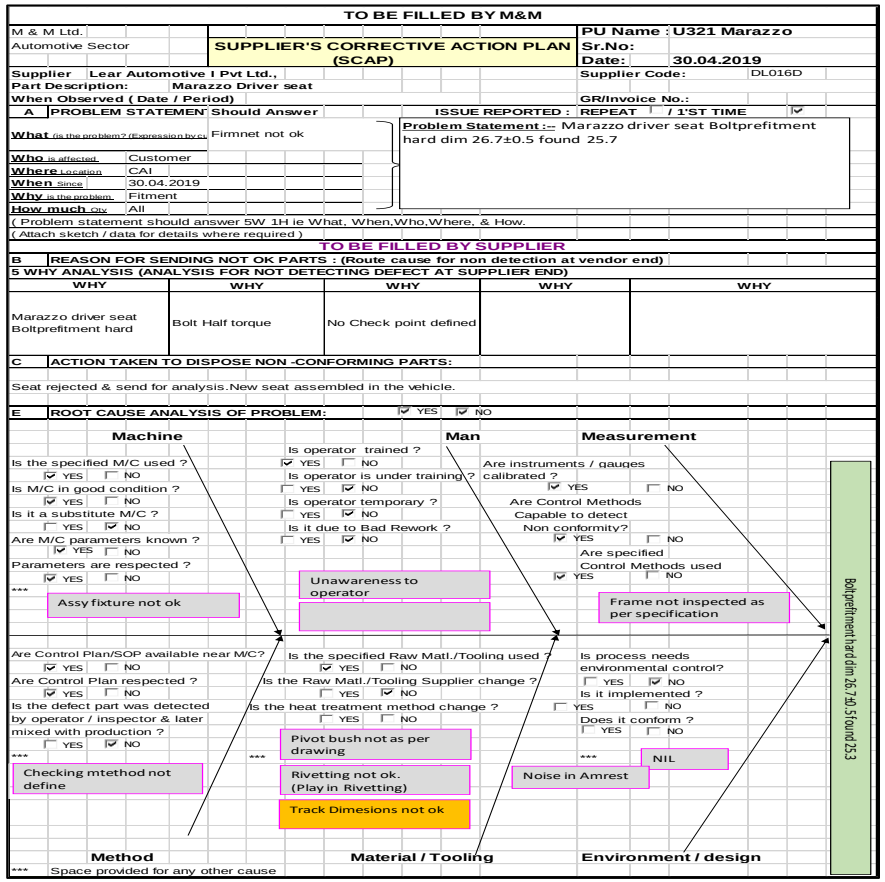
6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Actual date	Responsibility	Status
11	Seat bracket and body alignment not okay	Track riveting fixture block at supplier end to be replaced.	30.08.2019	Kiran Badgujar	CLOSED

OBSERVATIONS AND SUPPLIER CORRECTIVE ACTION PLAN (SCAP) :-

- Track riveting fixture block worn out.
- Track riveting fixture block replaced. Dimension dim 26.7±0.5 checked & found 25.67 & is ok now.
- Track riveting fixture block check point added in CLITA/PM Check sheet.



TO BE FILLED BY M&M

M & M Ltd. Automotive Sector | SUPPLIER'S CORRECTIVE ACTION PLAN (SCAP) | PU Name : U321 Marazzo | Sr.No: | Date: 30.04.2019

Supplier: Lear Automotive I Pvt Ltd., | Supplier Code: DL016D
Part Description: Marazzo Driver seat
When Observed (Date / Period) | GR/Invoice No.:

A PROBLEM STATEMENT Should Answer | ISSUE REPORTED : REPEAT / 1'ST TIME

What (is the problem? (Expression by cu): Firmnet not ok
Who (is affected): Customer
Where (Location): CAI
When (Since): 30.04.2019
Why (is the problem): Fitment
How much (Qty): All

Problem Statement :-- Marazzo driver seat Boltprefitment hard dim 26.7±0.5 found 25.7

(Problem statement should answer 5W 1H ie What, When,Who,Where, & How.
(Attach sketch / data for details where required)

TO BE FILLED BY SUPPLIER

B REASON FOR SENDING NOT OK PARTS : (Route cause for non detection at vendor end)
5 WHY ANALYSIS (ANALYSIS FOR NOT DETECTING DEFECT AT SUPPLIER END)

WHY	WHY	WHY	WHY	WHY
Marazzo driver seat Boltprefitment hard	Bolt Half torque	No Check point defined		

C ACTION TAKEN TO DISPOSE NON -CONFORMING PARTS:
Seat rejected & send for analysis.New seat assembled in the vehicle.

E ROOT CAUSE ANALYSIS OF PROBLEM: YES NO

Machine: Is the specified M/C used ? YES NO; Is M/C in good condition ? YES NO; Is it a substitute M/C ? YES NO; Are M/C parameters known ? YES NO; Parameters are respected ? YES NO; *** Assy fixture not ok

Man: Is operator trained ? YES NO; Is operator is under training ? YES NO; Is operator temporary ? YES NO; Is it due to Bad Rework ? YES NO; Unawareness to operator

Measurement: Are instruments / gauges calibrated ? YES NO; Are Control Methods Capable to detect Non conformity? YES NO; Are specified Control Methods used YES NO; Frame not inspected as per specification

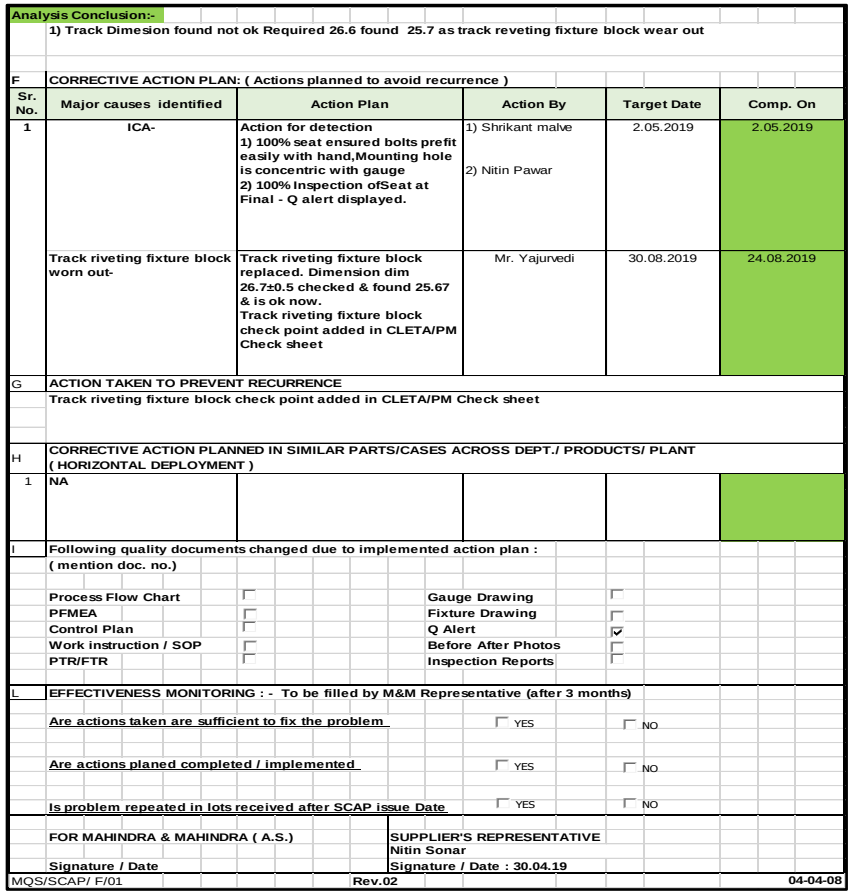
Method: Are Control Plan/SOP available near M/C? YES NO; Are Control Plan respected ? YES NO; Is the defect part was detected by operator / inspector & later mixed with production ? YES NO; *** Checking mtethod not define

Material / Tooling: Is the specified Raw Matl./Tooling used ? YES NO; Is the Raw Matl./Tooling Supplier change ? YES NO; Is the heat treatment method change ? YES NO; *** Pivot bush not as per drawing; Rivetting not ok. (Play in Rivetting); Track Dimesions not ok

Environment / design: Is process needs environmental control? YES NO; Is it implemented ? YES NO; Does it conform ? YES NO; *** NIL; Noise in Amrest

*** Space provided for any other cause

Boltprefitment hard dim 26.7±0.5 found 25.3



Analysis Conclusion:-
1) Track Dimesion found not ok Required 26.6 found 25.7 as track reveting fixture block wear out

F CORRECTIVE ACTION PLAN: (Actions planned to avoid recurrence)

Sr. No.	Major causes identified	Action Plan	Action By	Target Date	Comp. On
1	ICA-	Action for detection 1) 100% seat ensured bolts prefit easily with hand,Mounting hole is concentric with gauge 2) 100% Inspection ofSeat at Final - Q alert displayed.	1) Shrikant malve 2) Nitin Pawar	2.05.2019	2.05.2019
	Track riveting fixture block worn out-	Track riveting fixture block replaced. Dimension dim 26.7±0.5 checked & found 25.67 & is ok now. Track riveting fixture block check point added in CLETA/PM Check sheet	Mr. Yajurvedi	30.08.2019	24.08.2019

G ACTION TAKEN TO PREVENT RECURRENCE
Track riveting fixture block check point added in CLETA/PM Check sheet

H CORRECTIVE ACTION PLANNED IN SIMILAR PARTS/CASES ACROSS DEPT./ PRODUCTS/ PLANT (HORIZONTAL DEPLOYMENT)

1	NA				

I Following quality documents changed due to implemented action plan : (mention doc. no.)

Process Flow Chart ☐ | Gauge Drawing ☐
PFMEA ☐ | Fixture Drawing ☐
Control Plan ☐ | Q Alert ☑
Work instruction / SOP ☐ | Before After Photos ☐
PTR/FTR ☐ | Inspection Reports ☐

L EFFECTIVENESS MONITORING : - To be filled by M&M Representative (after 3 months)

Are actions taken are sufficient to fix the problem ☐ YES ☐ NO
Are actions planed completed / implemented ☐ YES ☐ NO
Is problem repeated in lots received after SCAP issue Date ☐ YES ☐ NO

FOR MAHINDRA & MAHINDRA (A.S.) | SUPPLIER'S REPRESENTATIVE Nitin Sonar
Signature / Date | Signature / Date : 30.04.19

MQS/SCAP/ F/01 Rev.02 04-04-08

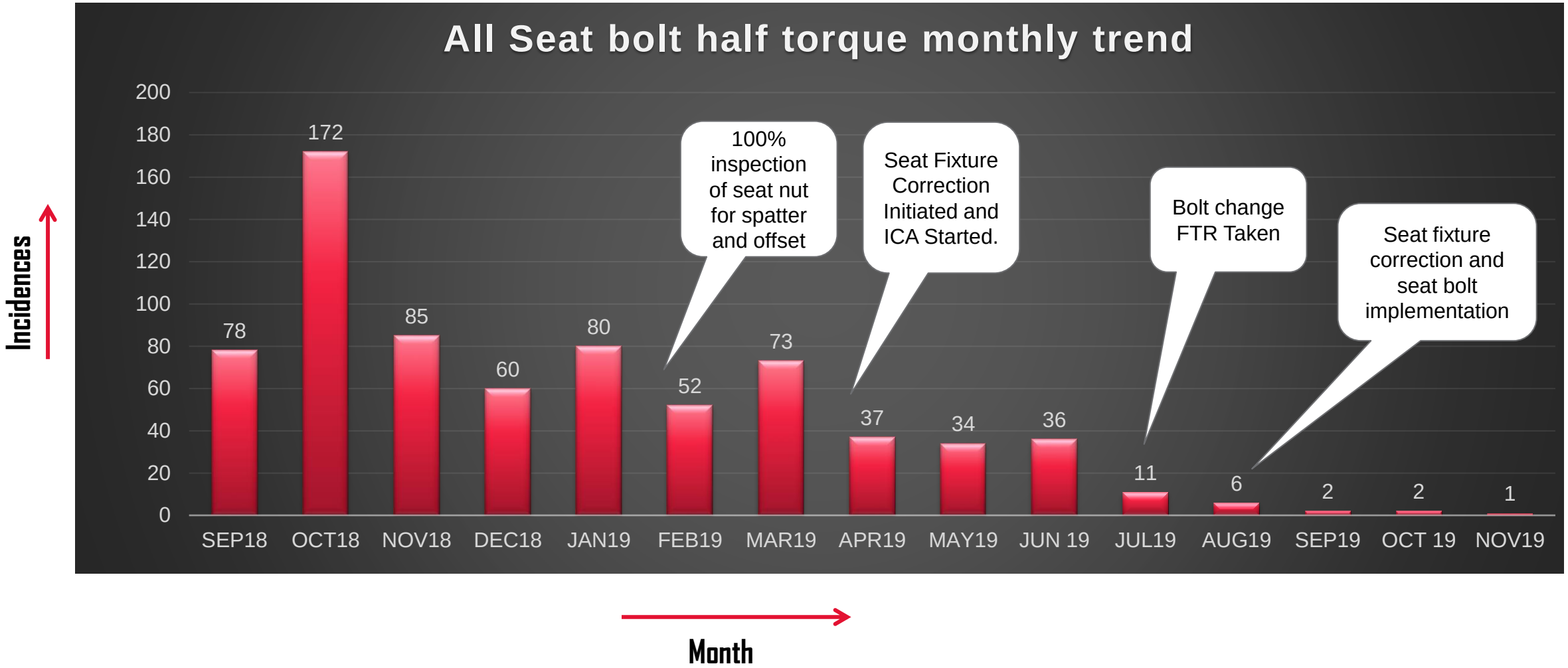
7. Check & confirm the results & effects

7.1 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
4	Weld nut Plate offset	20.02.2019	Keshav Badane	CLOSED
7	EN for bolt end change-Dock point to be added.	23.07.2019	Arvind Mistry	CLOSED
9	Check point added at Body Buy-off.	20.02.2019	Rahul Deore	CLOSED
11	SCAP for seat bracket match with body hole.	24.08.2019	Kiran Badjugar	CLOSED

7. Check & confirm the results & effects

7.2 Verify Results:



7. Check & confirm the results & effects

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Elimination of seat bolt half torque..
2) 58065 Rs Cost Saved/ Annum
3) FRC improved by 1% from 0 %
4) Repeat and chronic concern bolt scrappage eliminated.

In tangible Benefits :-

1) Improved Customer Satisfaction.
2) Major probable field issue protected to pass on.
3) Operator fatigue reduced , moral improved.

RE: COPQ for MYB project.

DR DHANOKAR RAJESH
To NAGE SHREEYASH

Ok approved

From: NAGE SHREEYASH
Sent: Sunday, February 16, 2020 7:25 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: COPQ for MYB project.

Dear Sir,
Please approve COPQ cost for MYB Project.

Rework Cost estimation for seat bolt half torque-		
Description	Units	
Rework time (Approx)	Min	20
Avg Concerns/Day	Nos	3
Total Required Rework time per day	Min	60
Effective working time	Min	465
Operator required /day	nos	0.13
Approx Operator cost/ day	Rs	1500
Operator cost /concern	Rs	193.55
Avg no. of working days	days	25
Operator cost /month	Rs	4838.71
Operator Cost/year	**Rs**	**58064.52**

Total Rework cost	Rs / Year	58065

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	EN for screw change.	07.08.2019	Arvind Mistry	CLOSED
2	FTR , DCR, Add delete.	23.07.2019	Arvind Mistry	CLOSED
3	New screw drawing	23.07.2019	Arvind Mistry	CLOSED
4	Control plan and Q alert at supplier end for seat fixture.	24.08.2019	Kiran Badgujar	CLOSED

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1	Training with SOS to line associates	23.08.2019	Dnyaneshwar Wagh.	Closed

8.3 Additional Audit & check

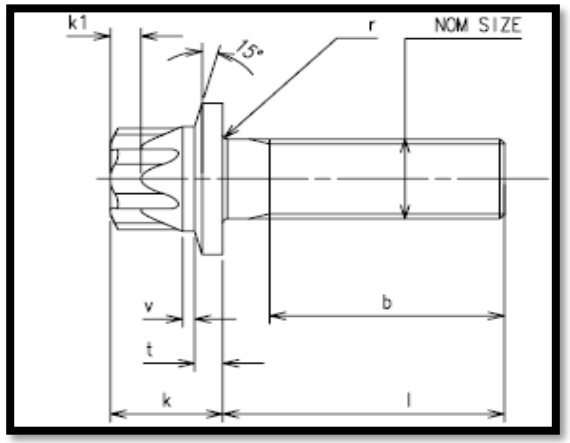
1	Check point added in body floor check-man check-sheet.	24.02.2019	Rahul Deore	Closed

8. Standardization

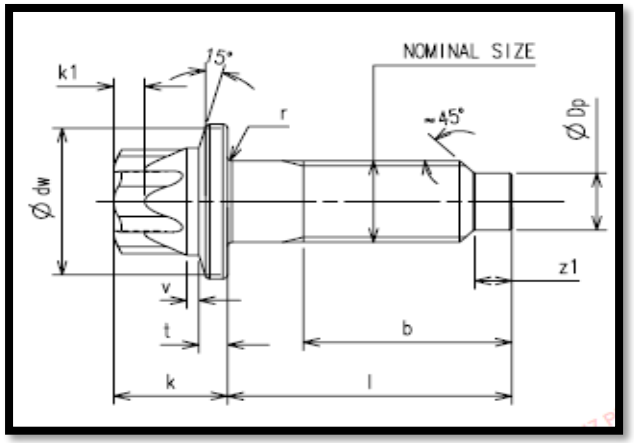
8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
7	FTR , DCR , Add delete	23.07.2019	Arvind Mistry	CLOSED

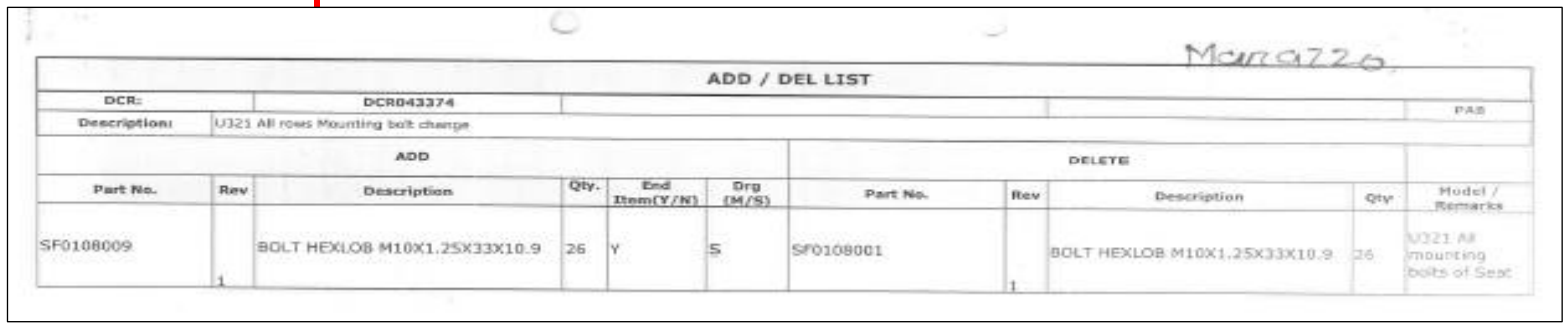
BEFORE



AFTER



EN NO: U321-EN-A003-000133.



ADD / DEL LIST

DCR:	DCR043374									PAB
Description:	U321 All rows Mounting bolt change									
ADD						DELETE				
Part No.	Rev	Description	Qty.	End Item(Y/N)	Drg (M/S)	Part No.	Rev	Description	Qty	Model / Remarks
SF0108009	1	BOLT HEXLOB M10X1.25X33X10.9	26	Y	S	SF0108001	1	BOLT HEXLOB M10X1.25X33X10.9	26	U321 All mounting bolts of Seat

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
7	FTR , DCR.	23.07.2019	Arvind Mistry	CLOSED

Mahindra & Mahindra Ltd.

Mahindra Rise. DESIGN CHANGE REQUEST

DCR Number	DCR044240
Source Of Change Request	202475 - RAUT AVESH
Product Owner	202475 - RAUT AVESH
Sub Product Owner	211714 - PATHAK YOGESH
Location	AD-NASIK
Reason Code	Quality Improvement
Change Description	ADDITION OF DOCK POINT ON ALL SEAT MOUNTING BOLTS
Why this change ?	TO AVOID BOLT SLIPPAGE ISSUE DURING FITMENT
Proposed Change	ADDITION OF DOCK POINT ON ALL SEAT MOUNTING BOLTS
Priority	Normal
CAE Required	No
CAE	
DFMEA/QFD Required	Yes
Designer	23075413 - LAHIRE MILIND
Lifecycle Status	Approved

Estimated Cost Impact

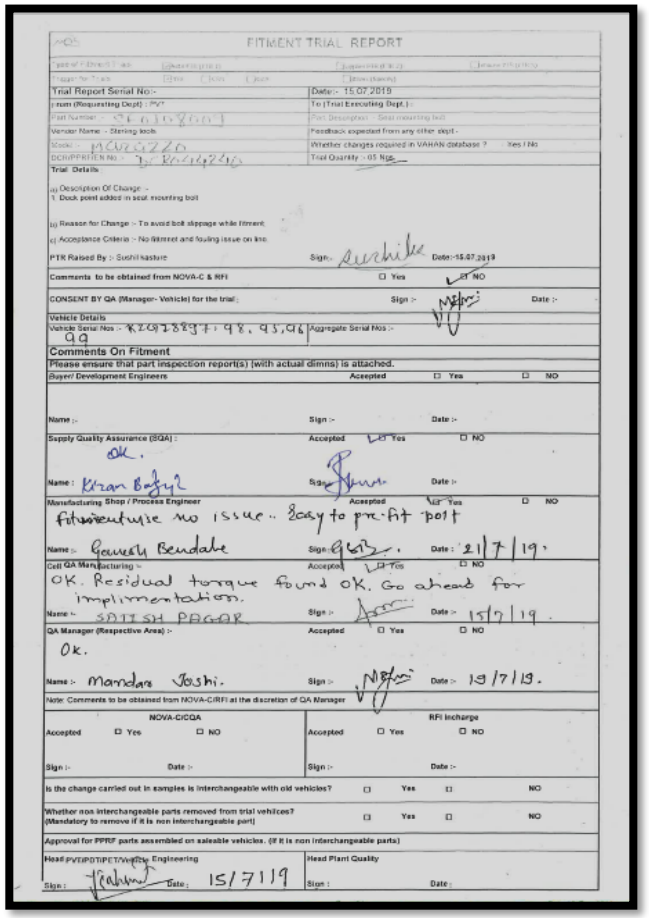
Type	Saving	Increase
Material		
Conversion		
ManPower		

Affected Model/Major Agg List

MARAZZO ALL

Approval Stage	Approver	Role	Status	Approval Date	Remarks
Initiation	23075413 - LAHIRE MILIND	Initiator	Initiated	22-JUL-2019	
Preliminary Approval by SPO	211714 - PATHAK YOGESH	SPO	Approved	23-JUL-2019	
Work Assigned to DFMEA/QFD	23075413 - LAHIRE MILIND	DFMEA	Completed	23-JUL-2019	DFMEA UPDATED
DFMEA/QFD Review by SPO	211714 - PATHAK YOGESH	SPO	Approved	23-JUL-2019	
ECN Release	23075413 - LAHIRE MILIND	DESIGNER	In Progress		

Jul 23, 2019, 3:23 PM



FITMENT TRIAL REPORT

Trial Report Serial No :- Date :- 15.07.2019

Trial Details

PTR Raised By :- Sushil kasture Date:-15.07.2019

Comments to be obtained from NOVA-C & RFI ☐ Yes ☑ NO

CONSENT BY QA (Manager- Vehicle) for the trial : Date :-

Vehicle Details

Comments On Fitment

Please ensure that part inspection report(s) (with actual dimns) is attached.

Buyer/ Development Engineers Accepted ☐ Yes ☐ NO

Supply Quality Assurance (SQA) : Accepted ☑ Yes ☐ NO

Manufacturing Shop / Process Engineer Accepted ☑ Yes ☐ NO

Fitment wise no issue.. Easy to pre-fit part

Name :- Ganesh Bendale Date : 21/7/19

Cell QA Manufacturing :- Accepted ☑ Yes ☐ NO

OK. Residual torque found OK. Go ahead for implimentation.

Name :- SATISH PAGAR Date :- 15/7/19.

QA Manager (Respective Area) :- Accepted ☐ Yes ☐ NO

Ok.

Name :- Mandar Joshi. Date :- 19/7/19.

Note: Comments to be obtained from NOVA-C/RFI at the discretion of QA Manager

NOVA-C/CQA Accepted ☐ Yes ☐ NO Sign :- Date :-

RFI Incharge Accepted ☐ Yes ☐ NO Sign :- Date :-

Is the change carried out in samples is interchangeable with old vehicles? ☐ Yes ☐ NO

Whether non interchangeable parts removed from trial vehicles? (Mandatory to remove if it is non interchangeable part) ☐ Yes ☐ NO

Approval for PPRF parts assembled on saleable vehicles. (If it is non interchangeable parts)

Head PV/EP/DTIPET/Vehicle Engineering Sign : Date : 15/7/19

Head Plant Quality Sign : Date :

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
2	EN for bolt change.	07.08.2019	Arvind Mistry	CLOSED

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise

Engineering Notice Release: U321-EN-A003-000133
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date 29-Jul-2019
Initiated Date 02-Aug-2019

Engineering Notice No : U321-EN-A003-000133

Released Date : 07-Aug-2019 11:17

Reference Change Req No : DCR044240

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : ADDITION OF DOCK POINT ON SEAT MOUNTING BOLTS (SF0108009) TO AVOID BOLT SLIP DURING FITMENT

Reason : ADDITION OF DOCK POINT ON SEAT MOUNTING BOLTS (SF0108009) TO AVOID BOLT SLIP DURING FITMENT

Quantity : -(Regular)

Reason Code : Quality Improvement

Priority : Mandatory

Note : ADDITION OF DOCK POINT ON SEAT MOUNTING BOLTS (SF0108009) TO AVOID BOLT SLIP DURING FITMENT

Affected Models : AU32BPES8T151A00SK, AU32CPES8T151A00SK, AU32EPES8T151A00SK, AU32EPES7T151A00SK, AU32FPES7T151A00SK, AU32BPES8T151A00XH, AU32CPES8T151A00XH, AU32EPES8T151A00XH, AU32EPES7T151A00XH, AU32FPES7T151A00XH, AU32BPES8T151A00WP, AU32CPES8T151A00WP, AU32EPES8T151A00WP, AU32EPES7T151A00WP, AU32FPES7T151A00WP, AU32BPES8T151A00GR, AU32CPES8T151A00GR, AU32EPES8T151A00GR, AU32EPES7T151A00GR, AU32FPES7T151A00GR, AU32BPES8T151A00PD, AU32CPES8T151A00PD, AU32EPES8T151A00PD, AU32EPES7T151A00PD, AU32FPES7T151A00PD, AU32BPES8T151A00ME, AU32CPES8T151A00ME, AU32EPES8T151A00ME, AU32EPES7T151A00ME, AU32FPES7T151A00ME, AU32CPES7T151A00GR, AU32CPES7T151A00SK, AU32BPES7T151A00SK, AU32BPES7T151A00GR, AU32BPES7T151A00PD, AU32BPES7T151A00ME, AU32CPES7T151A00WP, AU32BPES7T151A00XH, AU32CPES7T151A00ME, AU32CPES7T151A00PD, AU32CPES7T151A00XH, AU32FPES8T151A00SK, AU32FPES8T151A00PD, AU32FPES8T151A00ME, AU32FPES8T151A00XH

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : Bipin Melinkeri (208477)

Engineer : Bipin Melinkeri (208477)

Approved By : 211714-PATHAK YOGESH

Authorized By : HEAD-PVT-AD_202475-RAUT AVESH

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

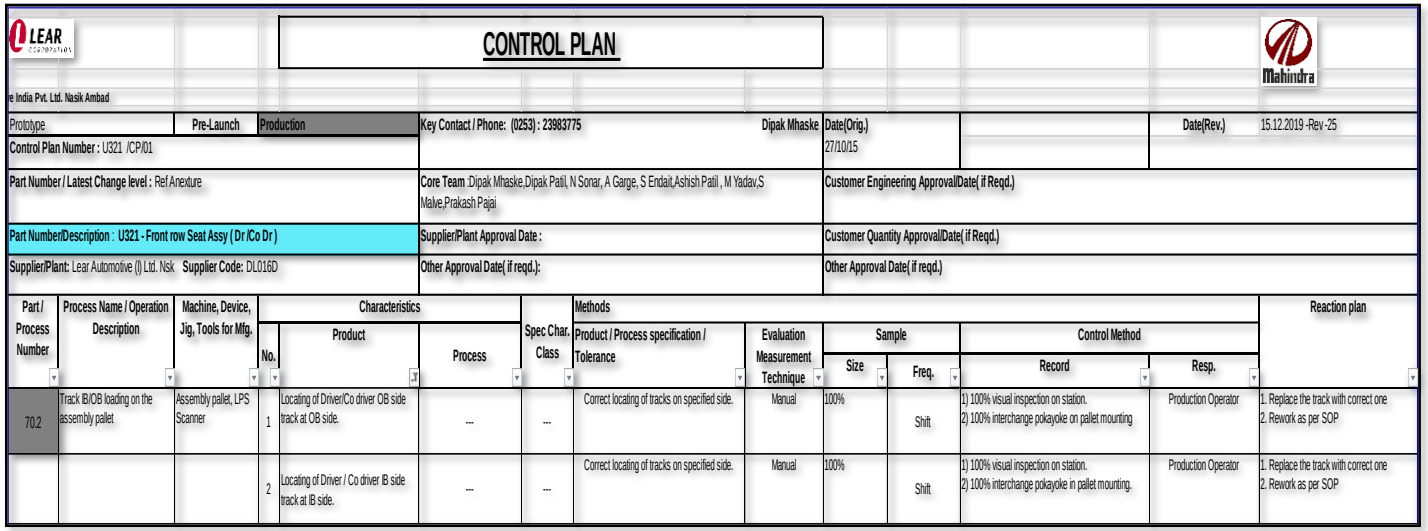
Added/Deleted Qty : It's delta of existing released EN for the same parent part number

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
11	Control Plan and Q alert for seat fixture at vendor end.	24.08.2019	Kiran Badgujar	CLOSED



LEAR CORPORATION

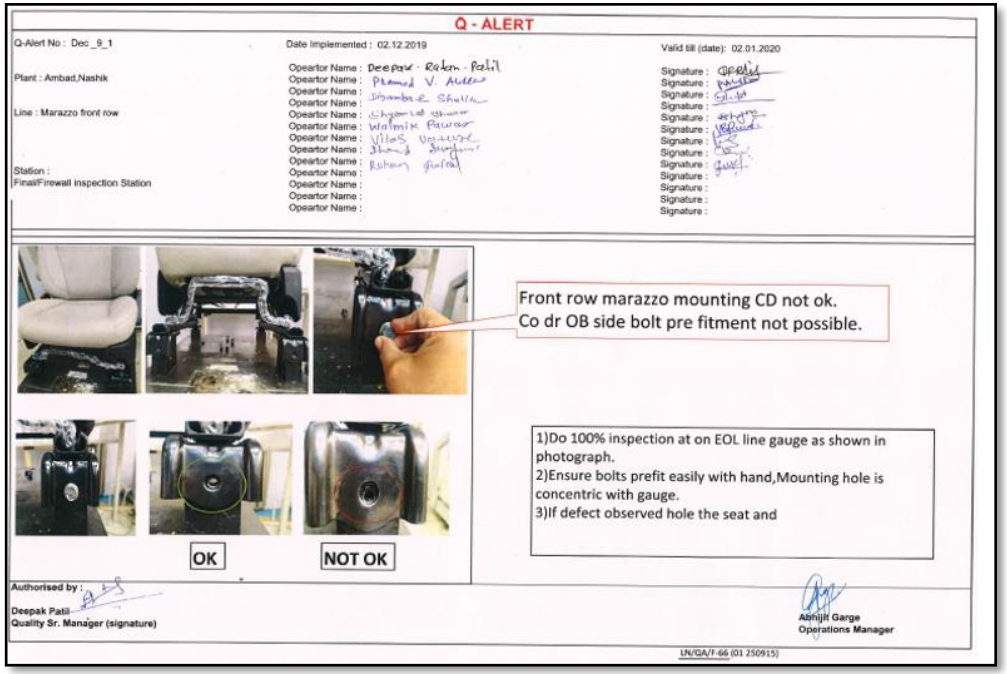
e India Pvt. Ltd. Nasik Ambad

CONTROL PLAN

Mahindra

Prototype	Pre-Launch	Production	Key Contact / Phone: (0253) : 23983775 Dipak Mhaske	Date(Orig.) 27/10/15	Date(Rev.)	15.12.2019 -Rev -25
Control Plan Number : U321 /CP/01						
Part Number / Latest Change level : Ref Anexture			Core Team :Dipak Mhaske,Dipak Patil, N Sonar, A Garge, S Endait,Ashish Patil , M Yadav,S Malve,Prakash Pajai	Customer Engineering Approval/Date(if Reqd.)		
Part Number/Description : U321 - Front row Seat Assy (Dr /Co Dr)			Supplier/Plant Approval Date :	Customer Quantity Approval/Date(if Reqd.)		
Supplier/Plant: Lear Automotive (I) Ltd. Nsk Supplier Code: DL016D			Other Approval Date(if reqd.):	Other Approval Date(if reqd.)		

Part / Process Number	Process Name / Operation Description	Machine, Device, Jig, Tools for Mfg.	Characteristics			Spec Char. Class	Methods						Reaction plan
			No.	Product	Process		Product / Process specification / Tolerance	Evaluation Measurement Technique	Sample		Control Method		
									Size	Freq.	Record	Resp.	
70.2	Track IB/OB loading on the assembly pallet	Assembly pallet, LPS Scanner	1	Locating of Driver/Co driver OB side track at OB side.	---	---	Correct locating of tracks on specified side.	Manual	100%	Shift	1) 100% visual inspection on station. 2) 100% interchange pokayoke on pallet mounting	Production Operator	1. Replace the track with correct one 2. Rework as per SOP
			2	Locating of Driver / Co driver IB side track at IB side.	---	---	Correct locating of tracks on specified side.	Manual	100%	Shift	1) 100% visual inspection on station. 2) 100% interchange pokayoke in pallet mounting.	Production Operator	1. Replace the track with correct one 2. Rework as per SOP



Q - ALERT

Q-Alert No : Dec _9_1

Date Implemented : 02.12.2019

Valid till (date): 02.01.2020

Plant : Ambad,Nashik

Line : Marazzo front row

Station : Final/Firewall inspection Station

Opeartor Name : Deepak Raban Patil
Opeartor Name : Pramod V. Akkar
Opeartor Name : Jayambar Shukla
Opeartor Name : Shyamlal Yadav
Opeartor Name : Vaibhik Pawar
Opeartor Name : Vilas Vaghare
Opeartor Name : Dhanraj Surgane
Opeartor Name : Rubani Gurkh

Signature :

Front row marazzo mounting CD not ok.
Co dr OB side bolt pre fitment not possible.

OK

NOT OK

1)Do 100% inspection at on EOL line gauge as shown in photograph.
2)Ensure bolts prefit easily with hand,Mounting hole is concentric with gauge.
3)If defect observed hole the seat and

Authorised by :

Deepak Patil
Quality Sr. Manager (signature)

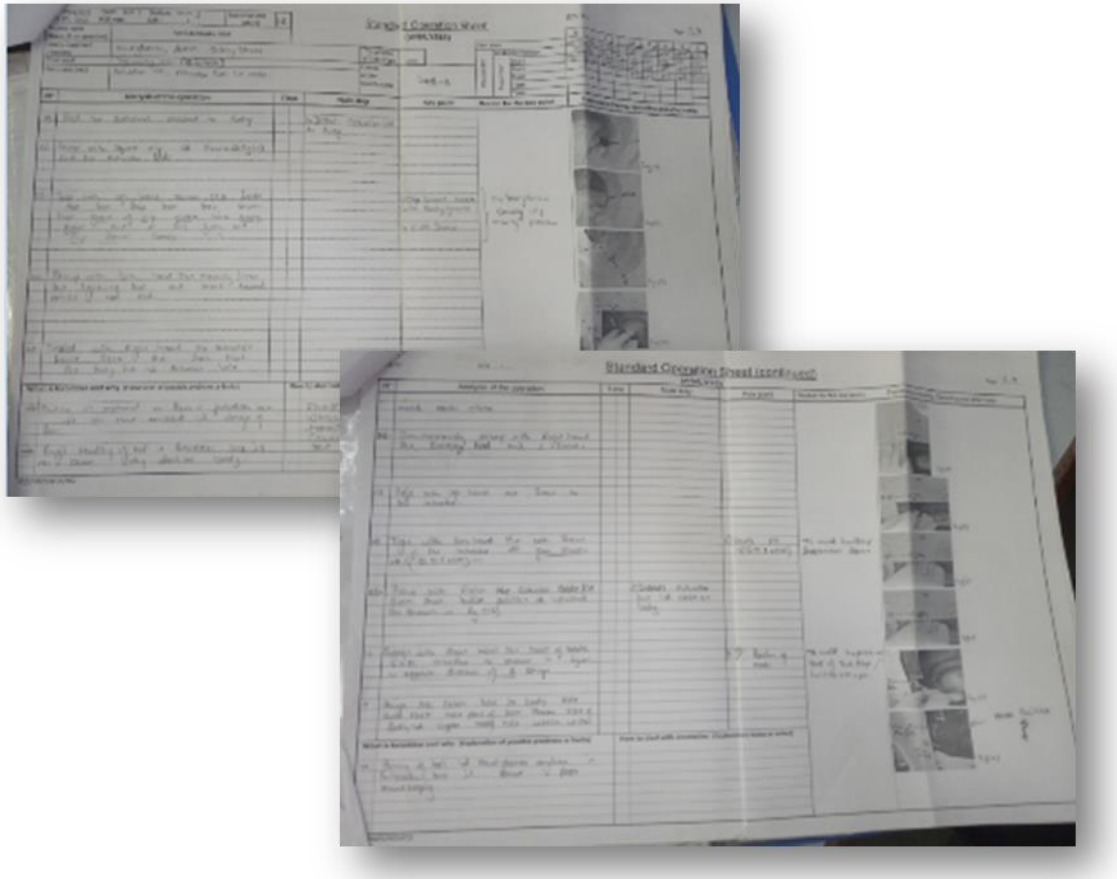
Abhijit Garge
Operations Manager

LN/QA/F-66 (01 250915)

8. Standardization

8.2 Training:

Action Plan	Target Dt	Resp.	Status
Training with SOS to line associates	23.08.2019	Dnyaneshwar Wagh.	Closed



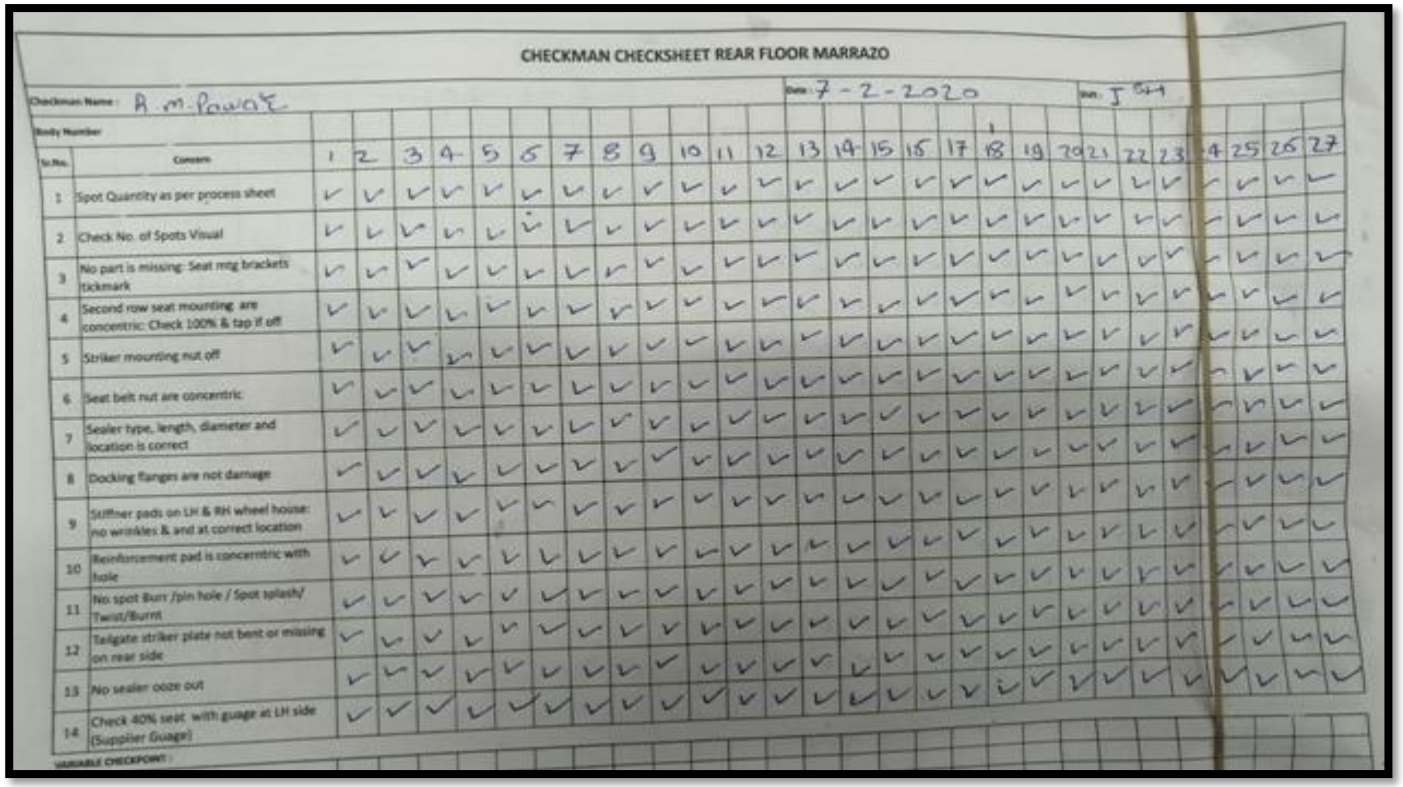
Stage associate and buy off inspectors are made aware about defect reported.

Detection and prone causes shared with line associate.

8. Standardization

8.3 Additional Audit & check .

SN	Document	Actual date	Responsibility	Status
9	Bodyshop Check-man check-sheet for body floor inspection updated for nut offset and welding spatter inside nut.	24.02.2019	Rahul Deore	CLOSED



9. Summarise & Plan future actions

9.1 Conclusion:

After taking corrective action in bolt design, modification of seat fixture, adding pin in nut before welding and adding checkpoint issue of seat bolt half torque is reduced more than 90% in SXTCF.

9.2 Horizontal Deployment:

Similar change implemented for KUV100 vehicle. Lesson learnt for upcoming projects of Thar.

9.3 Lessons Learnt:

Things Gone Wrong :

Horizontal deployment from Scorpio vehicle for seat bolt not done. Timely maintenance of fixture not updated in control plan of supplier.

Things Gone Right :

Design and fastener standard study always plays important role in problem solving.

Supplier recommendation taken into consideration while analysis, Gemba observations plays important role in problem solving.

Thank you

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